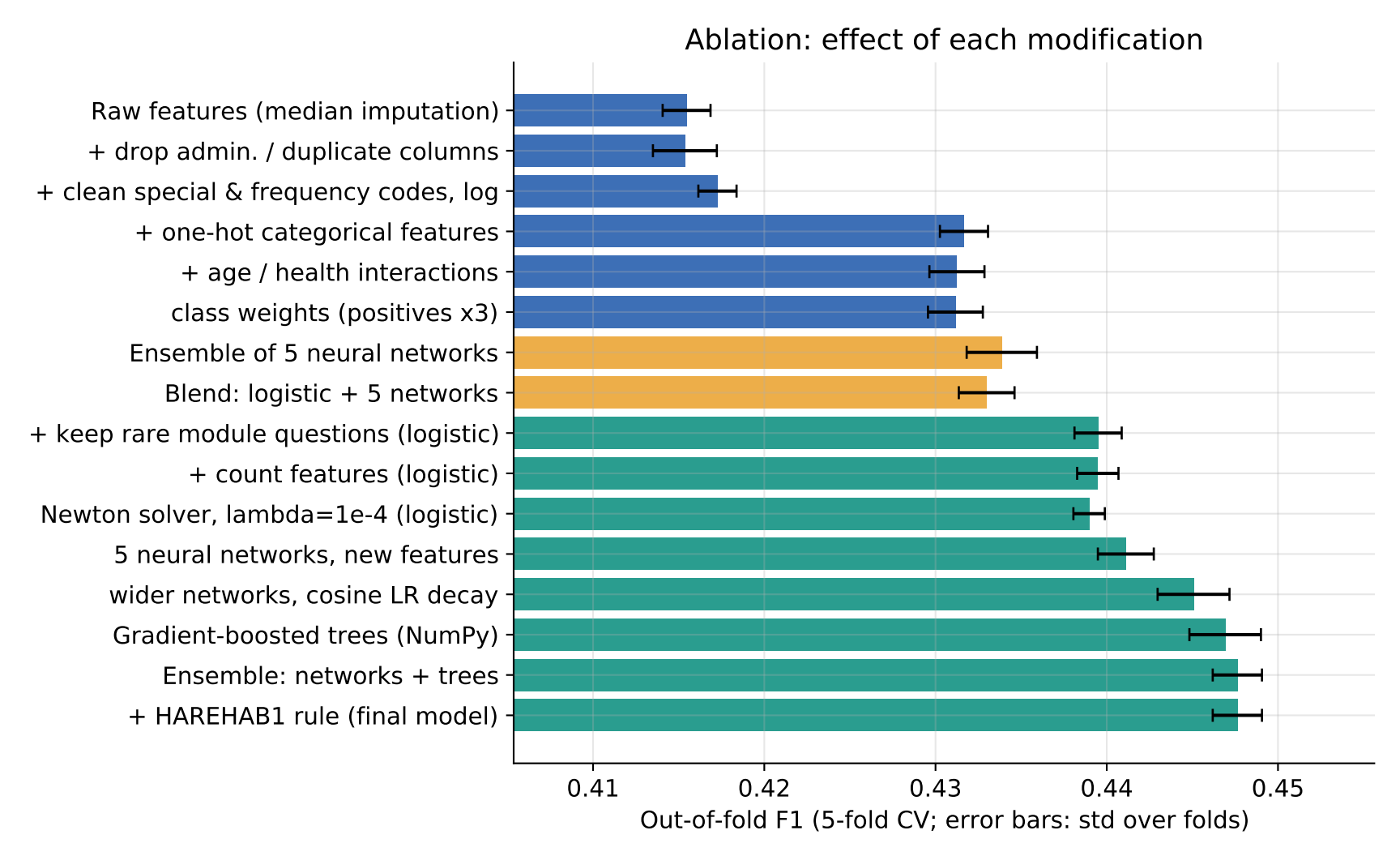
Ablation: effect of each modification
Raw features (median imputation)
+ drop admin. / duplicate columns
+ clean special & frequency codes, log
+ one-hot categorical features
+ age / health interactions
class weights (positives x3)
Ensemble of 5 neural networks
Blend: logistic + 5 networks
+ keep rare module questions (logistic)
+ count features (logistic)
Newton solver, lambda=1e-4 (logistic)
5 neural networks, new features
wider networks, cosine LR decay
Gradient-boosted trees (NumPy)
Ensemble: networks + trees
+ HAREHAB1 rule (final model)
0.41
0.42
0.43
0.44
0.45
Out-of-fold F1 (5-fold CV; error bars: std over folds)